

EDUGENIE — CODING & SOLUTION

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KEY CODE SNIPPETS

1. SHARED AI HELPER (app/main.py)

```
def _gen(prompt: str) -> str:
    try:
        response = client.models.generate_content(
            model=MODEL_NAME, contents=prompt)
        return response.text.strip()
    except Exception as e:
        return f"API Error: {str(e)}"
```

2. Q&A ROUTE

```
@app.get("/qa")
async def qa(question: str):
    return {"answer": _gen(question)}
```

3. EXPLAIN ROUTE

```
@app.post("/explain")
async def explain(request: Request):
    data = await request.json()
```

```
return {"explanation": _gen(
    f"Explain {data.get('topic','')} in simple terms for a student.")}
```

4. SUMMARIZE ROUTE

```
@app.post("/summarize")
async def summarize(request: Request):
    data = await request.json()
    return {"summary": _gen(
        f"Summarize in simple language:\n\n{data.get('text','')}")}
```

5. QUIZ ROUTE (parses JSON from Gemini)

```
@app.post("/quiz")
async def quiz(request: Request):
    data = await request.json()
    prompt = f"""Create 3 MCQs from this passage.
    Return ONLY valid JSON:
    [{"question": "...", "options": ["A", "B", "C", "D"], "answer": "A"}]
    Passage: {data.get('text','')}"""
    raw = _gen(prompt)
    if raw.startswith("API Error"):
        return {"quiz": [{"error": raw}]}
    try:
        raw = re.sub(r"```json|```", "", raw).strip()
        return {"quiz": json.loads(raw)}
```

```
except Exception as e:
```

```
    return {"quiz": [{"error": str(e)}]}
```

6. LEARNING PATH ROUTE

```
@app.get("/learn/recommendations")
```

```
async def learn(topic: str):
```

```
    return {"topic": topic, "recommendation": _gen(
```

```
        f"Create a beginner-to-advanced learning path for {topic} with resources.")}
```

7. FRONTEND FETCH EXAMPLE (index.html)

```
function askQ() {
```

```
    const q = document.getElementById('q').value;
```

```
    fetch('/qa?question=' + encodeURIComponent(q))
```

```
        .then(r => r.json())
```

```
        .then(d => document.getElementById('qaResult').innerText = d.answer);
```

```
}
```